



Week 1 (May 19)

N.B.: This research journal is maintained by Asfi Hassan to document the progress, ideas, implementation, and findings arising from research conducted under the supervision of Dr. Ahmed Biniaz at the University of Windsor as part of the NSERC Undergraduate Student Research Award (USRA). This project investigates a computational geometry approach to a problem closely related to the Lonely Runner Problem, a well-known open problem in number theory and discrete mathematics. The purpose of this journal is to provide a clear and organized record of my work throughout the Summer 2026 research term. It is intended to document my understanding, progress, and software development rather than to claim original authorship of all ideas presented. Many of the mathematical concepts and techniques discussed are drawn from existing research and are included to support my learning and to show how they inform my investigations and implementations. Entries may include mathematical observations, algorithmic developments, software design and implementation details, computational experiments, visualizations, meeting notes, and reflections on challenges encountered during the project. This journal serves as a comprehensive record of both the research process and the software tools developed to better understand this problem. Artificial intelligence tools were used to assist in understanding advanced mathematical concepts and research papers, and to support the preparation and refinement of this research journal at an undergraduate computer science level.

Foundational Understanding of the Lonely Runner Problem

Review of Related Literature

The Lonely Runner Conjecture (LRC) is a well-known problem in number theory and combinatorics involving runners moving at distinct constant speeds around a unit circle. The conjecture states that for any set of $n + 1$ runners, each runner will eventually be at least $\frac{1}{n+1}$ units away from every other runner along the circle. By fixing one runner at speed 0, the problem reduces to finding a time t such that the minimum circular distance between the stationary runner and all moving runners is at least $\frac{1}{n+1}$. In this project, the problem is interpreted geometrically by plotting time on the x-axis and the corresponding distance function on the y-axis, producing periodic piecewise-linear graphs whose peaks are farthest from the stationary runner. This transformation converts an abstract modular arithmetic problem into a visual and computational geometry problem. The resulting graphs reveal repeating patterns and critical points where the minimum distance function reaches local maxima. These visual structures make it easier to identify candidate witness times and compare the behavior of different speed sets.

Several key papers guided this research. Ram Krishna Pandey proved the conjecture for lacunary speed sets using a constructive interval-based argument, while recent works by Matthieu Rosenfeld and by Touch Sungkawichai and Tanupat Trakulthongchai used computer-assisted methods to verify the conjecture for up to thirteen total runners. These papers show that the problem can be reduced to a finite search over integer speed sets using modular sieves, bounded reductions, and exhaustive computation. Additional recent preprints refine these techniques by combining modular arithmetic, combinatorial reasoning, and algorithmic filtering to eliminate large classes of potential counterexamples. Inspired by these ideas, this project develops an interactive Python framework using Tkinter and Matplotlib to visualize the distance functions and explore the algorithmic structure underlying one of the most significant open problems in modern combinatorics and number theory.

Problem Sketch from Week 1

During the first meeting in week 1 of the research term, I studied a series of introductory sketches provided by my supervisor, Dr. Ahmed Biniiaz, where he explained the core ideas behind the Lonely Runner Problem and several related concepts. These sketches used the unit circle to illustrate how runners move at different constant speeds and showed that a runner is considered lonely when they are at least $1/(k+1)$ units away from every other runner, where $k+1$ is the total number of runners. By fixing one runner with speed 0, the problem becomes easier to analyze geometrically, since the distance between a moving runner and the stationary runner increases up to a maximum of $1/2$ of the unit circle and then decreases symmetrically. This geometric interpretation provides an intuitive way to visualize the problem and forms the basis for representing and studying it using computational geometry techniques within a plane.

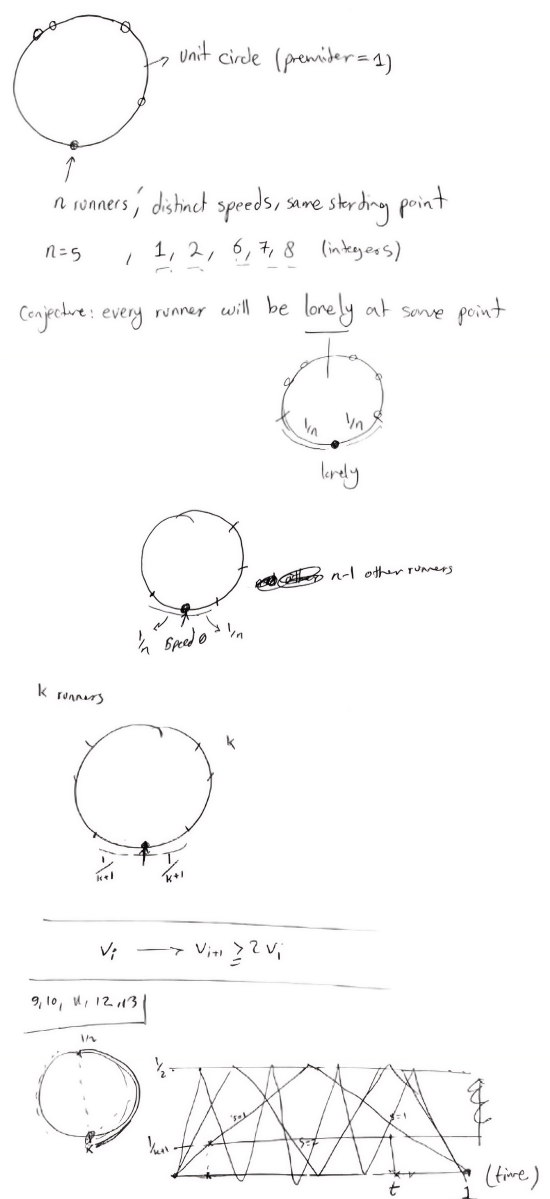


Figure 1: Introductory sketches illustrating the unit circle and graphical interpretation of the Lonely Runner Problem. Sketches provided by Dr. Ahmed Biniiaz.

Current Software Implementation

To support the geometric interpretation of the Lonely Runner Conjecture, an initial Python prototype was developed using the `Tkinter` and `Matplotlib` libraries. `Tkinter` is used to create a graphical user interface (GUI), while `Matplotlib` is used to generate and display mathematical plots directly within the application window. The program launches in full-screen mode and provides two input fields: one for the number of speeds, denoted by n , and another for the corresponding integer velocity values entered as a space-separated list.

The core functionality is implemented in the `draw()` function. After validating the user input, the program iterates through each velocity v and computes a sequence of equally spaced isosceles triangles over the interval $[0, 1]$. Each triangle has base length $\frac{1}{v}$ and height $\frac{1}{2}$, representing one period of the distance-to-nearest-integer function associated with that speed. Since the function $\|tv\|$ is periodic and piecewise linear, the repeated triangles provide an exact geometric representation of the runner's relative distance from the stationary runner over time. Different colors are assigned to distinguish the graphs corresponding to different speeds.

The application also includes a `reset()` function, which clears all input fields and removes the current plot, allowing new speed sets to be entered efficiently. The plotting area is embedded directly into the GUI using `FigureCanvasTkAgg`, and the aspect ratio is fixed so that all triangles retain their correct geometric proportions. This prototype establishes the foundational visualization framework for the project and serves as the basis for future extensions, including the computation of intersection points, automatic identification of candidate loneliness times, and algorithmic verification of specific speed configurations.

The core plotting algorithm iterates through each input velocity v and constructs its corresponding distance function. For each speed, the interval $[0, 1]$ is divided into v equal subintervals of length $\frac{1}{v}$. In each subinterval, the program computes the left endpoint, right endpoint, and midpoint, then draws an isosceles triangle of height $\frac{1}{2}$. Each triangle represents one period of the distance-to-nearest-integer function $\|tv\|$. Repeating this process over all subintervals produces the full piecewise-linear graph for that runner, with distinct colors used to differentiate multiple speeds. The code snippet is provided as a figure below:

```
for idx, v in enumerate(values):
    color = colors[idx % len(colors)]
    triangle_base = line_length / v

    for i in range(v):
        base_left_x = i * triangle_base
        base_right_x = (i + 1) * triangle_base
        top_vertex_x = (base_left_x + base_right_x) / 2

        ax.plot(
            [base_left_x, base_right_x, top_vertex_x, base_left_x],
            [0, 0, y, 0],
            color=color
        )
```

Figure 2: Core algorithm above used to generate the triangular representation of the conjecture.

Week 2 (May 26)

Review and Examination of Various Cases

During this weeks discussions with Dr. Ahmed Biniaz, we examined why the Lonely Runner Conjecture becomes significantly more difficult as the number of runners increases. In particular, we discussed the complexity of modern sieving techniques, which use filtering methods to eliminate impossible speed configurations during large computational searches. I was told to take a deeper read into *The Lonely Runner Conjecture turns 60*, to become familiar with concepts. Several special cases of the conjecture were also suggested for independent study and proof attempts, including weaker loneliness bounds and restricted families of runner speeds.

Case 1 — Proving the Weaker Bound $\frac{1}{2k}$

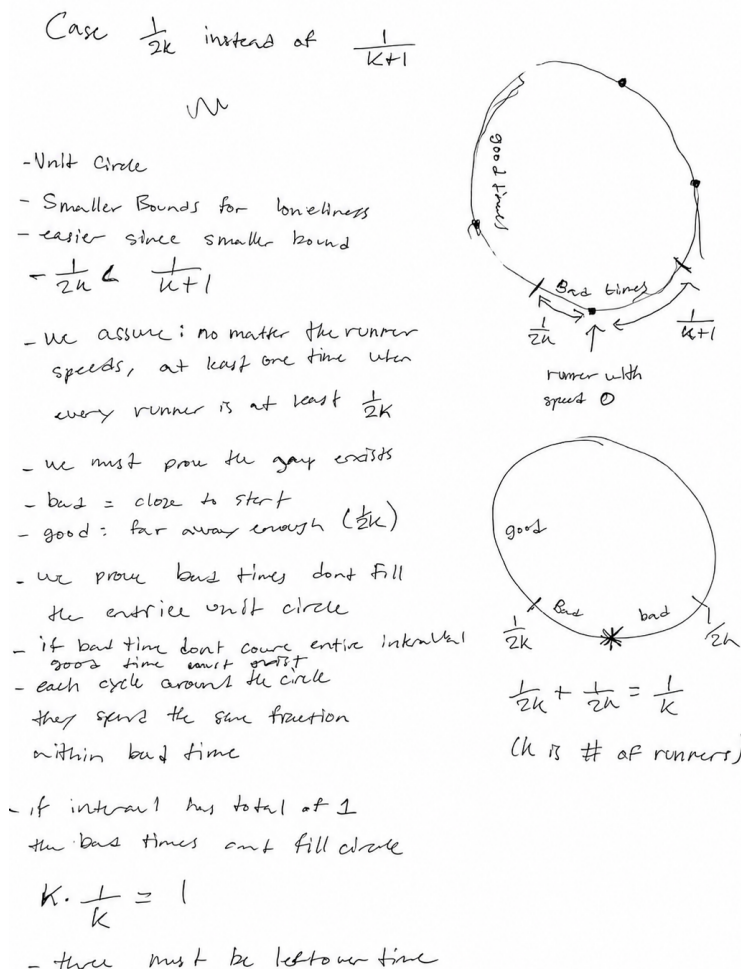


Figure 3: Illustration of the good and bad regions on the unit circle for the $1/(2k)$ loneliness bound and brainstorming proof for Case 1.

This proof establishes a weaker version of the Lonely Runner Conjecture by showing that there exists a time t such that every runner is at least $\frac{1}{2k}$ away from the starting point. Here, k represents the number of moving runners, and the runner speeds are denoted by $v_1, v_2, \dots, v_k$.

For each runner v_i , define the set of “bad times” as

$$A_i = \left\{ t \in (0, 1) : \|tv_i\| < \frac{1}{2k} \right\}.$$

The idea of “bad times” is simple: these are the times when a runner is still too close to the starting point, meaning the runner is not yet lonely. We create a set A_i to collect all of these bad times together for runner i . The notation $t \in (0, 1)$ means we only examine times between 0 and 1, since the runners move periodically around the unit circle and their positions repeat after one full cycle.

In this expression, t represents time, v_i is the speed of the i -th runner, and $\|tv_i\|$ represents the distance from the runner to the nearest integer position on the unit circle, which corresponds to the distance from the starting point. The condition

$$\|tv_i\| < \frac{1}{2k}$$

therefore means that runner i is closer than $\frac{1}{2k}$ to the starting point, making that time “bad.”

For a single runner, the total amount of bad time is less than $\frac{1}{k}$. Since there are k runners, the total amount of bad time across all runners is still less than the entire interval $(0, 1)$. Therefore, there must exist at least one time t which is not contained in any bad set A_i . At this time,

$$\|tv_i\| \geq \frac{1}{2k}$$

for every runner i . Hence, every runner is sufficiently far from the starting point simultaneously.

Weaker Bound $\frac{1}{2k}$, when $k = 4$

To understand the idea, consider 4 runners. In this case, each runner is considered “bad” when they are within $\frac{1}{8}$ of the starting point. Each runner is only bad for approximately one quarter of the time. At first glance, four runners each contributing one quarter of the interval appears to cover the entire timeline. However, many of these bad intervals overlap, meaning the same moments are counted multiple times. As a result, the bad times cannot cover the entire interval $(0, 1)$. Therefore, there must exist at least one time where none of the runners are bad, implying that all four runners are simultaneously at least $\frac{1}{8}$ away from the starting point.

This does not prove the full Lonely Runner Conjecture, which requires a distance of $\frac{1}{k+1}$, but it does establish the existence of a guaranteed weaker loneliness distance of $\frac{1}{2k}$.

Reference: Guillem Perarnau and Oriol Serra, *The Lonely Runner Conjecture turns 60*, Section 5: “Gap of loneliness.” The weaker bound $\kappa(n) \geq \frac{1}{2n}$ is attributed to Wills.

Case 2 — Speeds Growing by a Factor of Two

This case proves the Lonely Runner Conjecture when the runner speeds increase rapidly. Specifically, the speeds satisfy

$$v_{i+1} \geq 2v_i.$$

This means each runner moves at least twice as fast as the previous runner. For example, possible speeds could be 1, 2, 4, 8, 16. Such sequences are known as lacunary sequences.

The condition $v_{i+1} \geq 2v_i$ is what makes the proof work. Since each runner is moving at least twice as fast as the previous runner, the corresponding distance function repeats more frequently over the interval $[0, 1]$. Intuitively, this creates many opportunities to find times where additional runners satisfy the loneliness condition without eliminating all previously valid times.

The goal is to show that there exists a time t where every runner is at least

$$\frac{1}{k+1}$$

away from the starting point, where k is the number of moving runners.

The proof works by constructing nested intervals of valid times. An interval is simply a continuous range of times. For example, the interval $[0.2, 0.4]$ contains every time between 0.2 and 0.4. The notation $I_2 \subseteq I_1$ means that I_2 is completely contained inside I_1 .

First, an interval I_1 is chosen where the first runner is sufficiently far from the starting point. Then a smaller interval $I_2 \subseteq I_1$ is selected where both the first and second runners satisfy the loneliness condition. This process continues:

$$I_1 \supset I_2 \supset I_3 \supset \cdots \supset I_k.$$

Because each speed is at least twice as large as the previous one, the distance patterns repeat more frequently, creating enough opportunities to continue choosing smaller valid intervals. Since every interval is contained in the previous one, any time remaining in the final interval automatically works for all runners considered so far.

At the end of the construction, the final interval I_k is still nonempty, meaning it contains at least one valid time t . Therefore,

$$\|tv_i\| \geq \frac{1}{k+1}$$

for every runner i , and all runners become lonely simultaneously.

Reference: Ram Krishna Pandey, *A Note on the Lonely Runner Conjecture*. Also summarized in Guillem Perarnau and Oriol Serra, *The Lonely Runner Conjecture turns 60*, Section 7.3: “Lacunary sequences.”

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Case 3 — When $\frac{v_k}{v_1} \leq k$

This case proves the Lonely Runner Conjecture when the fastest runner is not excessively faster than the slowest runner. The condition is

$$\frac{v_k}{v_1} \leq k,$$

where v_1 is the smallest speed, v_k is the largest speed, and k is the number of moving runners. Equivalently,

$$v_k \leq kv_1.$$

This means the fastest runner is at most k times faster than the slowest runner. For example, if there are 4 runners, then the fastest runner can be at most 4 times faster than the slowest runner. The speeds are therefore relatively close together and do not vary too dramatically.

The importance of this condition is that the runners move at comparable rates. If one runner were significantly faster than all the others, the distance patterns would be more difficult to analyze. By keeping the speeds within a controlled range, the structure of the problem becomes much more predictable.

The proof begins by defining

$$N = v_1 + v_k$$

Here, N is simply the sum of the smallest and largest speeds. Under the assumption above, all runner speeds lie inside a controlled interval between

$$\frac{N}{k+1} \quad \text{and} \quad N - \frac{N}{k+1}.$$

In other words, every speed is guaranteed to lie within a specific range determined by the slowest and fastest runners. This prevents any speed from being unusually small or unusually large relative to the others.

Because the speeds are confined to this range, the resulting distance functions exhibit a more regular structure. This allows the authors to apply known results showing that there exists a time t such that

$$\|tv_i\| \geq \frac{1}{k+1}$$

for every runner i . Therefore, all runners become lonely simultaneously.

In simple terms, the proof shows that when the runner speeds are reasonably close together, the system is sufficiently well-behaved that the Lonely Runner Conjecture can be established directly.

Reference: Guillem Perarnau and Oriol Serra, *The Lonely Runner Conjecture turns 60*, Section 7.1: “Simple remarks and further reductions.” ““

Week 3 (June 2)

Further Overview and Discussion of Cases

During this Week 3 meeting, Dr. Ahmed Biniiaz walked me through the three special cases of the Lonely Runner Conjecture that I have been studying. We discussed the intuition behind each proof, focusing on why the arguments work and how they relate to the overall conjecture. We also revisited Case 3, where the fastest runner is not excessively faster than the slowest runner, and clarified several details of the proof. In particular, Dr. Biniiaz explained how choosing the time

$$g = \frac{1}{v_1 + v_k}$$

and substituting it into the distance function $\|gv_i\|$ helps show that all runners remain sufficiently far from the starting point. This provided a clearer understanding of why the condition $\frac{v_k}{v_1} \leq k$ guarantees the loneliness requirement and how the distance function is used within the proof.

Dr. Biniiaz suggested keeping the definition

$$\|x\| = \min\{x - \lfloor x \rfloor, \lceil x \rceil - x\}$$

in mind when studying the conjecture, since it represents the distance from a real number to the nearest integer. This notation appears throughout the Lonely Runner literature and is fundamental to expressing the loneliness condition. Understanding this definition should help provide additional intuition for Case 3 and the role of the distance function in the conjecture.

For the coming week, I will continue reading the relevant papers and studying additional cases, proofs, reductions, and properties related to the Lonely Runner Conjecture. My goal is to strengthen my understanding of the existing literature and develop a clearer picture of the different techniques that have been used to prove special cases of the conjecture.

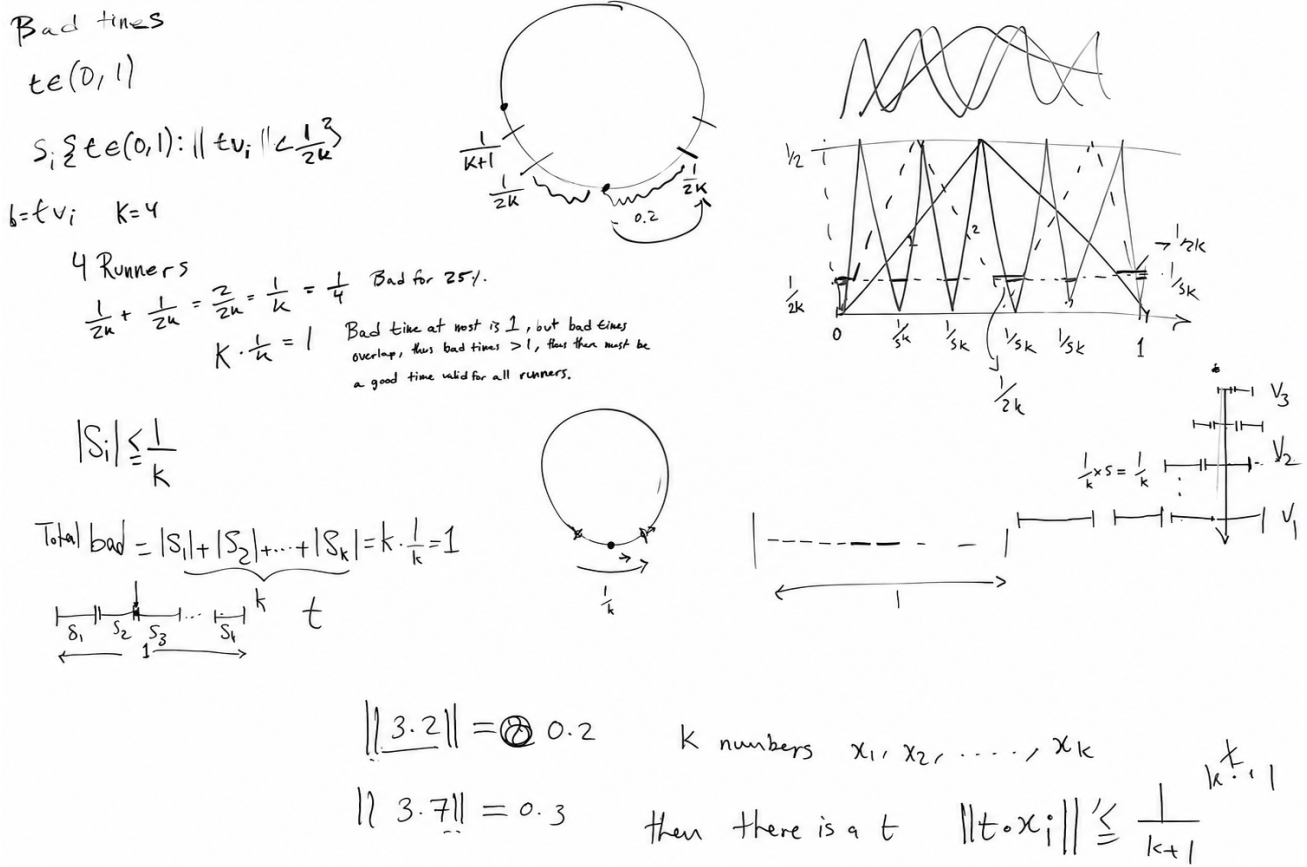


Figure 4: Sketches from meeting illustrating the main ideas behind Cases 1 and 2 of the Lonely Runner Conjecture, including bad times and runner separation on the unit circle.

Understanding the Distance Function $\|x\|$

The notation $\|x\|$ appears throughout the Lonely Runner literature and represents the distance from a real number x to the nearest integer. It is defined by

$$\|x\| = \min\{x - \lfloor x \rfloor, \lceil x \rceil - x\}.$$

To understand this formula, it is helpful to examine each symbol individually. The notation $\lfloor x \rfloor$ means the greatest integer less than or equal to x , often described as rounding x down. Similarly, $\lceil x \rceil$ means the smallest integer greater than or equal to x , often described as rounding x up. The notation $\min\{\cdot\}$ means that we select the smaller of the two values inside the braces.

The expression $x - \lfloor x \rfloor$ measures the distance from x to the integer immediately below it, while $\lceil x \rceil - x$ measures the distance from x to the integer immediately above it. Since $\|x\|$ represents the distance to the nearest integer, we simply take whichever of these two distances is smaller.

For example, if $x = 3.2$, then $\lfloor x \rfloor = 3$ and $\lceil x \rceil = 4$. The distance to 3 is 0.2, while the distance to 4 is 0.8. Since 0.2 is smaller, we obtain $\|3.2\| = 0.2$. Similarly, if $x = 3.8$, the distance to 3 is 0.8 and the distance to 4 is 0.2, so $\|3.8\| = 0.2$.

In the context of the Lonely Runner Conjecture, the quantity $\|tv_i\|$ measures how far runner i is from the starting point at time t . This distance function appears throughout the literature and is fundamental to expressing the loneliness condition. Understanding this notation provides important intuition for many proofs and reductions related to the conjecture.

Case 4 — Proof for 2 runners, where $n = 2$

For the case $n = 2$, the Lonely Runner Conjecture asks us to prove that

$$\kappa(2) = \frac{1}{3}.$$

Here, $n = 2$ means there are two moving speeds, usually written as

$$V = \{v_1, v_2\},$$

where $v_1 < v_2$. The notation $\kappa(V)$ represents the largest guaranteed loneliness distance for the speed set V . Therefore, proving $\kappa(2) = \frac{1}{3}$ means that for any two speeds, there is always some time when both moving runners are at least $\frac{1}{3}$ of a lap away from the stationary runner at the origin.

The main challenge is determining which time t to examine. Since time can take infinitely many values, checking every possible time individually would be impossible. The proof therefore looks for a structured family of candidate times that are likely to produce large distances from the starting point.

One important idea used in this proof is that the best loneliness time can be chosen in the form

$$t_0 = \frac{\ell}{v_1 + v_2},$$

where ℓ is a positive integer. At first glance, this formula may seem unusual, but its purpose is to balance the motion of the two runners. The quantity $v_1 + v_2$ is the combined speed of both runners, and using it in the denominator creates times that naturally align with the movement patterns of both runners simultaneously.

The symbol ℓ simply selects which candidate time we are testing. For example, the proof considers times such as

$$\frac{1}{v_1 + v_2}, \quad \frac{2}{v_1 + v_2}, \quad \frac{3}{v_1 + v_2},$$

and so on. Instead of searching through infinitely many real numbers, the proof only needs to study these special candidate times.

At such a time, the distance from runner i to the origin is measured using

$$\|t_0v_i\|.$$

Here, t_0v_i represents how far runner i has travelled after time t_0 . The notation $\|t_0v_i\|$ means the distance from that position to the nearest integer. Since every integer corresponds to completing a whole number

of laps and returning to the starting point, $\|t_0 v_i\|$ is exactly the distance from the runner to the starting point on the unit circle.

The proof then shows that one can choose a value of ℓ that places both runners as far from the starting point as possible. This leads to the bound

$$\kappa(V) = \frac{\lfloor (v_1 + v_2)/2 \rfloor}{v_1 + v_2}.$$

The symbol $\lfloor x \rfloor$ means rounding x down to the nearest integer. The quantity

$$\lfloor (v_1 + v_2)/2 \rfloor$$

therefore represents the largest whole number that does not exceed half of $v_1 + v_2$. Intuitively, this corresponds to placing the runners as close as possible to the halfway point around the circle, which maximizes their distance from the origin.

This fraction is always at least

$$\frac{1}{3},$$

which implies

$$\kappa(V) \geq \frac{1}{3}.$$

Therefore, for any two speeds v_1 and v_2 , there exists a time when both runners are at least $\frac{1}{3}$ of the track away from the origin.

In simple terms, the proof shows that with only two moving runners, their positions can always be balanced so that neither runner is too close to the starting point. This is why the Lonely Runner Conjecture is fully true for $n = 2$.

Reference: Guillem Perarnau and Oriol Serra, *The Lonely Runner Conjecture turns 60*, Section 6.1: “Case $n = 2$.”

Invisible Runners

The *Invisible Runner Problem* is a variation of the Lonely Runner Conjecture introduced by Czerwinski and Grytczuk. The main idea is simple: instead of requiring every runner to satisfy the loneliness condition, one runner is allowed to be ignored, or made “invisible.” The question then becomes whether the remaining runners can still achieve the required separation.

Suppose we have a set of runner speeds

$$V = \{v_1, v_2, \dots, v_n\},$$

where n is the number of runners and each v_i represents a speed. In the original Lonely Runner Conjecture, every runner must eventually become lonely. The Invisible Runner Problem asks what happens if we remove one runner from consideration and focus only on the remaining runners.

The main result states that for every set of n runner speeds, there is always at least one runner whose removal guarantees a loneliness distance of at least

$$\frac{1}{n}.$$

In mathematical notation, this is written as

$$\kappa(V \setminus \{v\}) \geq \frac{1}{n}.$$

Here, $V \setminus \{v\}$ means the speed set obtained after removing runner v , and κ represents the guaranteed loneliness distance. In simple terms, the theorem says that every speed set contains at least one runner whose removal makes the remaining problem easier.

The intuition behind the proof comes from averaging. Imagine observing all runners over time and counting how many runners are too close to the starting point. The proof shows that if the guaranteed loneliness distance were smaller than $\frac{1}{n}$, then the average number of runners that are too close to the origin would be less than 2.

This is important because if there were always at least two runners too close to the starting point, then the average would have to be at least 2. Since the average is actually less than 2, there must be some moment when there is at most one runner that is too close to the origin.

For example, suppose there are five runners. At a particular time, it might happen that four runners are already far enough from the starting point, while only one runner remains too close. At that moment, the loneliness condition is almost satisfied. The only obstacle is the single runner that is still too close to the origin.

This observation is the key idea behind the Invisible Runner Problem. If we simply remove the one runner causing the problem and make that runner “invisible,” then all of the remaining runners already satisfy the required distance of at least $\frac{1}{n}$ from the starting point. Therefore, there must always exist at least one runner whose removal guarantees the loneliness condition for the rest of the group.

Reference: Guillem Perarnau and Oriol Serra, *The Lonely Runner Conjecture turns 60*, Section 10.1: “Invisible runners.”

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